

09/21/2026: Set operations

CSCI 246: Discrete Structures

Textbook reference: Sec. 12, Scheinerman

Today's Agenda

- Review exercises (≈ 10 mins)
- Mini-lecture (≈ 25 mins)
- Group exercises (≈ 15 mins)

Subtopics from today's reading

Scheinerman Sec. 12 (Set Operations) covers the following subtopics

1. Union and intersection
 - a. Introduction
 - b. Properties
 - c. The size of a union
2. Set Difference and Symmetric Difference
3. Cartesian Product

Outline for Mini-Lecture:

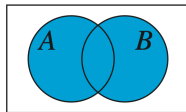
- **Union and Intersection**
 - **Introduction**
 - **Properties**
 - **Size of a Union**
- **Set & Symmetric Difference**
- **Cartesian Product**

Union and Intersection

Let A and B be sets.

Definition. The **union** of A and B , denoted $A \cup B$ is the set of all elements that are in A or B (or both).

$$A \cup B = \{x : x \in A \text{ or } x \in B\}$$

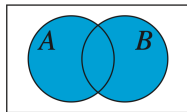


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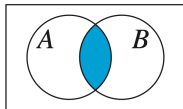
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Definition. The **intersection** of A and B , denoted $A \cap B$ is the set of all elements that are in both A and B .

$$A \cap B = \{x : x \in A \text{ and } x \in B\}$$



Union and Intersection: Emoji Math

$$F = \left\{ \boxed{\text{😊}}, \boxed{\text{😡}} \right\}$$

face emojis

$$N = \left\{ \boxed{\text{😡}}, \boxed{\text{💩}} \right\}$$

negative emojis

Union and Intersection: Emoji Math

$$F = \left\{ \img alt="Smiling Face with Smiling Eyes emoji" data-bbox="261 238 341 347"}, \img alt="Angry Face emoji" data-bbox="354 238 434 347" \right\}$$

face emojis

$$N = \left\{ \img alt="Angry Face emoji" data-bbox="261 363 341 471"}, \img alt="Poop emoji" data-bbox="354 363 434 471" \right\}$$

negative emojis

$$F \cap N = \left\{ \img alt="Angry Face emoji" data-bbox="261 498 341 597" \right\}$$

negative face emojis

Union and Intersection: Emoji Math

$$F = \left\{ \img alt="Smiling face with open mouth" data-bbox="268 258 341 331" , \img alt="Angry face" data-bbox="366 258 439 331" \right\}$$

face emojis

$$N = \left\{ \img alt="Angry face" data-bbox="268 384 341 457" , \img alt="Smiling poop" data-bbox="366 384 439 457" \right\}$$

negative emojis

$$F \cap N = \left\{ \img alt="Angry face" data-bbox="268 508 341 581" \right\}$$

negative face emojis

$$F \cup N = \left\{ \img alt="Smiling face with open mouth" data-bbox="268 631 341 704" , \img alt="Angry face" data-bbox="366 631 439 704" , \img alt="Smiling poop" data-bbox="464 631 537 704" \right\}$$

face or negative emojis

Outline for Mini-Lecture:

- **Union and Intersection**
 - Introduction
 - **Properties**
 - Size of a Union
- Set & Symmetric Difference
- Cartesian Product

Union and Intersection Properties

Overview. Set operations have many properties. E.g. the **distributive properties** are:

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C), \quad A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

Intuition. Let's apply the lefthand property to wind emojis W , face emojis F , and negative emojis N .

$$\begin{aligned} W \cup (F \cap N) &= \{ \text{🌬️}, \text{👉} \} \cup \left(\{ \text{😡}, \text{😄} \} \cap \{ \text{😡}, \text{💩} \} \right) \\ &= \{ \text{🌬️}, \text{👉} \} \cup \{ \text{😡} \} \\ &= \{ \text{🌬️}, \text{👉}, \text{😡} \} \\ &= \{ \text{🌬️}, \text{👉}, \text{😡}, \text{😄} \} \cap \{ \text{🌬️}, \text{👉}, \text{😡}, \text{💩} \} \\ &= \left(\{ \text{🌬️}, \text{👉} \} \cup \{ \text{😡}, \text{😄} \} \right) \cap \left(\{ \text{🌬️}, \text{👉} \} \cup \{ \text{😡}, \text{💩} \} \right) \\ &= (W \cup F) \cap (W \cup N) \end{aligned}$$

Union and Intersection Properties

Intuition. We can also build intuition about the set properties by creating an application.

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Suppose we work for Rosauers and we are studying how likely customers are to make a purchase in different parts of the store. Let



- P : The set of Rosauers customers who made a purchase.
- R : The set of Rosauers customers who were shopping for roses.
- C : The set of Rosauers customers who were shopping for candy.

Then

$$(R \cup C) \cap P \underbrace{=}_{\text{commutative}} P \cap (R \cup C) \underbrace{=}_{\text{distributive}} (P \cap R) \cup (P \cap C).$$

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Poll. What does this equation say?

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Poll. What does this equation say? The set of Rosauers customers who were shopping for roses or candy and made a purchase is the same as the set of Rosauers customers who were shopping for roses and made a purchase combined with the set of Rosauers customers who were shopping for candy and made a purchase.

Union and Intersection Properties: Proofs

Poll. How would we go about **proving** a set property – e.g. the first distributive property?

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

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Strategy. Recall that the definitions of union and intersection depends upon logical operators **and** and **or**

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Thus, we can rely upon the analogous properties from **Boolean algebra** to do all the heavy lifting.

- $x \wedge y = y \wedge x$ and $x \vee y = y \vee x$. (Commutative properties)
- $(x \wedge y) \wedge z = x \wedge (y \wedge z)$ and $(x \vee y) \vee z = x \vee (y \vee z)$. (Associative properties)
- $x \wedge \text{TRUE} = x$ and $x \vee \text{FALSE} = x$. (Identity elements)
- $\neg(\neg x) = x$.
- $x \wedge x = x$ and $x \vee x = x$.
- $x \wedge (y \vee z) = (x \wedge y) \vee (x \wedge z)$ and $x \vee (y \wedge z) = (x \vee y) \wedge (x \vee z)$. (Distributive properties)
- $x \wedge (\neg x) = \text{FALSE}$ and $x \vee (\neg x) = \text{TRUE}$.
- $\neg(x \wedge y) = (\neg x) \vee (\neg y)$ and $\neg(x \vee y) = (\neg x) \wedge (\neg y)$. (DeMorgan's Laws)

Do you remember the Boolean Algebra props. (Theorem 7.2)?!

Union and Intersection Properties: Proofs

Problem. The distributive properties for sets are

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C), \quad \text{and} \quad A \cap (B \cup C) = (A \cap B) \cup (A \cap C).$$

Prove that the first identity above holds.

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$$\begin{aligned} A \cup (B \cap C) &= \{x : x \in A \text{ or } x \in B \cap C\} && \text{(by def. union } \cup) \\ &= \{x : x \in A \text{ or } (x \in B \text{ and } x \in C)\} && \text{(by def. intersection } \cap) \end{aligned}$$

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Remark. We used the *distributive property for Boolean Algebra* (from Theorem 7.2) to prove the distributive property for sets.

Union and Intersection Properties: Visual Intuition

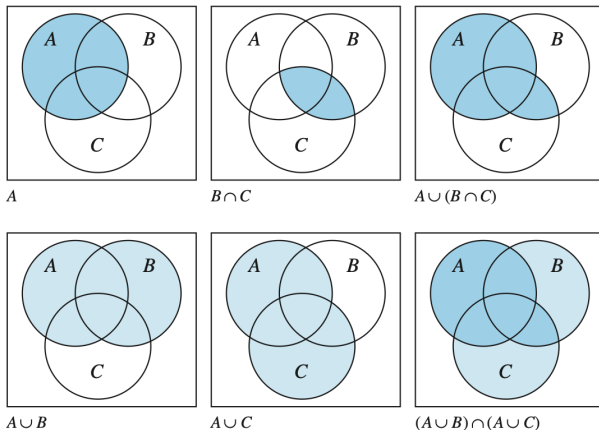
Venn diagrams are also useful for understanding set properties.

Union and Intersection Properties: Visual Intuition

Venn diagrams are also useful for understanding set properties.

Poll. How does the diagram below illustrate the distributive property

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$



Union and Intersection Properties

A complete set of properties related to set union and intersections is given in the text.

Theorem 12.3 Let A , B , and C denote sets. The following are true:

- $A \cup B = B \cup A$ and $A \cap B = B \cap A$. (Commutative properties)
 - $A \cup (B \cup C) = (A \cup B) \cup C$ and $A \cap (B \cap C) = (A \cap B) \cap C$. (Associative properties)
 - $A \cup \emptyset = A$ and $A \cap \emptyset = \emptyset$.
 - $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ and $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$. (Distributive properties)
-

Outline for Mini-Lecture:

- **Union and Intersection**
 - Introduction
 - Properties
 - **Size of a Union**
- Set & Symmetric Difference
- Cartesian Product

The Size of a Union

Poll. Discuss what $|A \cup B| = |A| + |B|$ means. Is it correct?

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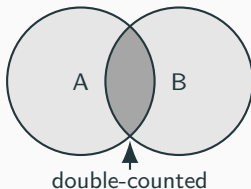
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Poll. Why?

The Size of a Union (Scheinerman Prop 12.4)

Let A and B be finite sets. Then

$$|A \cup B| = |A| + |B| - |A \cap B|$$



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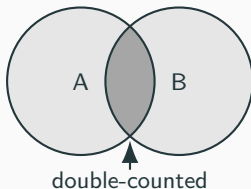
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Poll. When is this equation useful?

The Size of a Union

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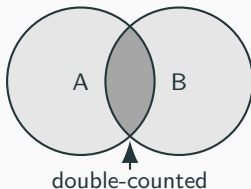
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Poll. When is this equation useful? **Solution.** When computing $|A|, |B|, |A \cap B|$ is easier than $|A \cup B|$.

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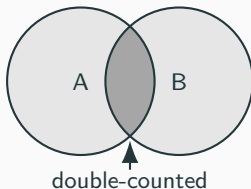
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Poll. When is this equation useful? **Solution.** When computing $|A|, |B|, |A \cap B|$ is easier than $|A \cup B|$. (Example: Today's Group Exercise #4.)

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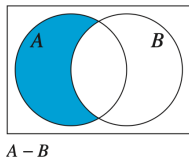
- Union and Intersection
- Set & Symmetric Difference
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Set and Symmetric Difference

Let A and B be sets.

Definition. The **set difference** of A and B , denoted $A - B$ is the set of all elements that are in A but not in B .

$$A - B = \{x : x \in A \text{ and } x \notin B\}$$

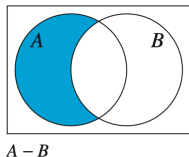


Set and Symmetric Difference

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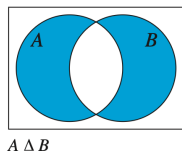
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Definition. The **symmetric difference** of A and B , denoted $A \triangle B$ is the set of all elements that are in A but not B or B but not A .

$$A \triangle B = (A - B) \cup (B - A)$$



Set and Symmetric Difference: Emoji Math

$$F = \left\{ \boxed{\text{😊}}, \boxed{\text{😡}} \right\} \quad \text{face emojis}$$

$$N = \left\{ \boxed{\text{😡}}, \boxed{\text{💩}} \right\} \quad \text{negative emojis}$$

Set and Symmetric Difference: Emoji Math

$$F = \left\{ \img alt="happy face emoji" data-bbox="218 288 291 361" , \img alt="angry face emoji" data-bbox="308 288 381 361" \right\} \quad \text{face emojis}$$

$$N = \left\{ \img alt="angry face emoji" data-bbox="218 412 291 485" , \img alt="poop emoji" data-bbox="308 412 381 485" \right\} \quad \text{negative emojis}$$

$$F - N = \left\{ \img alt="happy face emoji" data-bbox="218 538 291 611" \right\} \quad \text{happy face emojis}$$

Set and Symmetric Difference: Emoji Math

$$F = \left\{ \img alt="happy face emoji" data-bbox="218 288 291 361" , \img alt="angry face emoji" data-bbox="311 288 384 361" \right\} \quad \text{face emojis}$$

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$$F - N = \left\{ \img alt="happy face emoji" data-bbox="218 539 291 612" \right\} \quad \text{happy face emojis}$$

$$F \triangle N = \left\{ \img alt="happy face emoji" data-bbox="218 662 291 735" , \img alt="poop emoji" data-bbox="311 662 384 735" \right\} \quad \text{non-negative face and non-face negative emojis}$$

Symmetric Difference: Alternate Characterization

A characterization of symmetric difference (Scheinerman Prop 12.11)

Let A and B be sets. Then

$$A \triangle B = (A \cup B) - (A \cap B)$$

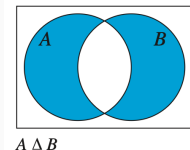
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Visual Intuition.



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Poll. How could we prove this is true?

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Reminder from Sec. 10: Proving two sets are equal

To show $S = T$, we show $S \subseteq T$ and $T \subseteq S$.

- $S \subseteq T$. We show that if $x \in S$, then $x \in T$.
- $T \subseteq S$. We show that if $x \in T$, then $x \in S$.

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To show $S = T$, we show $S \subseteq T$ and $T \subseteq S$.

- $S \subseteq T$. We show that if $x \in S$, then $x \in T$. In this case, we show that if $x \in A \triangle B$, then $x \in (A \cup B) - (A \cap B)$.
- $T \subseteq S$. We show that if $x \in T$, then $x \in S$.

Remark. For details, see the textbook.

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A characterization of symmetric difference (Scheinerman Prop 12.11)

Let A and B be sets. Then

$$A \triangle B = (A \cup B) - (A \cap B)$$

Poll. How could we prove this is true?

Reminder from Sec. 10: Proving two sets are equal

To show $S = T$, we show $S \subseteq T$ and $T \subseteq S$.

- $S \subseteq T$. We show that if $x \in S$, then $x \in T$. In this case, we show that if $x \in A \triangle B$, then $x \in (A \cup B) - (A \cap B)$.
- $T \subseteq S$. We show that if $x \in T$, then $x \in S$. In this case, we show that if $x \in (A \cup B) - (A \cap B)$, then $x \in A \triangle B$.

Remark. For details, see the textbook.

DeMorgan's Law

The textbook gives DeMorgan's Law as

Proposition 12.12 (DeMorgan's Laws) Let A , B , and C be sets. Then

$$A - (B \cup C) = (A - B) \cap (A - C) \quad \text{and} \quad A - (B \cap C) = (A - B) \cup (A - C).$$

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Remark. A simpler way to express this law is as follows. Let X be a “universe”; that is, every element we consider is in X . For any set S , define its **complement** $\bar{S}$ as points not in S (but in X). Then we can write DeMorgan's Law as

$$\overline{A \cup B} = \bar{A} \cap \bar{B}, \quad \overline{A \cap B} = \bar{A} \cup \bar{B}$$

DeMorgan's Law

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Remark. A simpler way to express this law is as follows. Let X be a “universe”; that is, every element we consider is in X . For any set S , define its **complement** $\bar{S}$ as points not in S (but in X). Then we can write DeMorgan's Law as

$$\overline{A \cup B} = \bar{A} \cap \bar{B}, \quad \overline{A \cap B} = \bar{A} \cup \bar{B}$$

That is,

- The complement of the union of two sets is the same as the intersection of their complements.
- The complement of the intersection of two sets is the same as the union of their complements

Outline for Mini-Lecture:

- Union and Intersection
- Set & Symmetric Difference
- Cartesian Product

Cartesian Product

Let A and B be sets.

Definition. The **Cartesian product** of A and B , denoted $A \times B$ is the set of all ordered pairs (two element lists) formed by taking an element of A together with an element from B in all possible ways.

$$A \times B = \{(a, b) : a \in A, b \in B\}$$

Cartesian Product: Emoji Math

$$F = \left\{ \begin{array}{|c|} \hline \img alt="Smiling Face with Smiling Eyes emoji" data-bbox="298 181 368 261"/> \\ \hline \end{array} , \begin{array}{|c|} \hline \img alt="Angry Face emoji" data-bbox="391 181 461 261"/> \\ \hline \end{array} \right\}$$

face emojis

$$N = \left\{ \begin{array}{|c|} \hline \img alt="Angry Face emoji" data-bbox="298 305 368 385"/> \\ \hline \end{array} , \begin{array}{|c|} \hline \img alt="Poop emoji" data-bbox="391 305 461 385"/> \\ \hline \end{array} \right\}$$

negative emojis

Cartesian Product: Emoji Math

$$F = \left\{ \begin{array}{|c|} \hline \img alt="Smiling face emoji" data-bbox="298 188 368 268"/> \\ \hline \end{array}, \begin{array}{|c|} \hline \img alt="Angry face emoji" data-bbox="398 188 468 268"/> \\ \hline \end{array} \right\}$$

face emojis

$$N = \left\{ \begin{array}{|c|} \hline \img alt="Angry face emoji" data-bbox="298 311 368 391"/> \\ \hline \end{array}, \begin{array}{|c|} \hline \img alt="Poop emoji" data-bbox="398 311 468 391"/> \\ \hline \end{array} \right\}$$

negative emojis

$$F \times N = \left\{ \left(\begin{array}{|c|} \hline \img alt="Smiling face emoji" data-bbox="353 478 433 558"/> \\ \hline \end{array}, \begin{array}{|c|} \hline \img alt="Angry face emoji" data-bbox="453 478 533 558"/> \\ \hline \end{array} \right), \left(\begin{array}{|c|} \hline \img alt="Smiling face emoji" data-bbox="593 478 673 558"/> \\ \hline \end{array}, \begin{array}{|c|} \hline \img alt="Poop emoji" data-bbox="693 478 773 558"/> \\ \hline \end{array} \right), \right. \\ \left. \left(\begin{array}{|c|} \hline \img alt="Angry face emoji" data-bbox="353 603 433 683"/> \\ \hline \end{array}, \begin{array}{|c|} \hline \img alt="Angry face emoji" data-bbox="453 603 533 683"/> \\ \hline \end{array} \right), \left(\begin{array}{|c|} \hline \img alt="Angry face emoji" data-bbox="593 603 673 683"/> \\ \hline \end{array}, \begin{array}{|c|} \hline \img alt="Poop emoji" data-bbox="693 603 773 683"/> \\ \hline \end{array} \right) \right\}$$

Cartesian Product: Emoji Math

$$F = \left\{ \begin{array}{|c|} \hline \img alt="Smiling face emoji" data-bbox="298 188 368 268"/> \\ \hline \end{array}, \begin{array}{|c|} \hline \img alt="Angry face emoji" data-bbox="398 188 468 268"/> \\ \hline \end{array} \right\}$$

face emojis

$$N = \left\{ \begin{array}{|c|} \hline \img alt="Angry face emoji" data-bbox="298 312 368 392"/> \\ \hline \end{array}, \begin{array}{|c|} \hline \img alt="Poop emoji" data-bbox="398 312 468 392"/> \\ \hline \end{array} \right\}$$

negative emojis

$$F \times N = \left\{ \left(\begin{array}{|c|} \hline \img alt="Smiling face emoji" data-bbox="358 482 428 562"/> \\ \hline \end{array}, \begin{array}{|c|} \hline \img alt="Angry face emoji" data-bbox="458 482 528 562"/> \\ \hline \end{array} \right), \left(\begin{array}{|c|} \hline \img alt="Smiling face emoji" data-bbox="598 482 668 562"/> \\ \hline \end{array}, \begin{array}{|c|} \hline \img alt="Poop emoji" data-bbox="698 482 768 562"/> \\ \hline \end{array} \right), \right. \\ \left. \left(\begin{array}{|c|} \hline \img alt="Angry face emoji" data-bbox="358 602 428 682"/> \\ \hline \end{array}, \begin{array}{|c|} \hline \img alt="Angry face emoji" data-bbox="458 602 528 682"/> \\ \hline \end{array} \right), \left(\begin{array}{|c|} \hline \img alt="Angry face emoji" data-bbox="598 602 668 682"/> \\ \hline \end{array}, \begin{array}{|c|} \hline \img alt="Poop emoji" data-bbox="698 602 768 682"/> \\ \hline \end{array} \right) \right\}$$

So $F \times N$ is a face emoji paired with a negative emoji.

Cartesian Product: Emoji Math

$$F = \left\{ \begin{array}{|c|} \hline \img alt="Smiling face with smiling eyes emoji" data-bbox="298 185 368 265"/> \\ \hline \end{array}, \begin{array}{|c|} \hline \img alt="Angry face emoji" data-bbox="388 185 458 265"/> \\ \hline \end{array} \right\}$$

face emojis

$$N = \left\{ \begin{array}{|c|} \hline \img alt="Angry face emoji" data-bbox="298 308 368 388"/> \\ \hline \end{array}, \begin{array}{|c|} \hline \img alt="Poop emoji" data-bbox="388 308 458 388"/> \\ \hline \end{array} \right\}$$

negative emojis

$$F \times N = \left\{ \left(\begin{array}{|c|} \hline \img alt="Smiling face with smiling eyes emoji" data-bbox="353 475 433 555"/> \\ \hline \end{array}, \begin{array}{|c|} \hline \img alt="Angry face emoji" data-bbox="453 475 523 555"/> \\ \hline \end{array} \right), \left(\begin{array}{|c|} \hline \img alt="Smiling face with smiling eyes emoji" data-bbox="593 475 673 555"/> \\ \hline \end{array}, \begin{array}{|c|} \hline \img alt="Poop emoji" data-bbox="693 475 763 555"/> \\ \hline \end{array} \right), \right. \\ \left. \left(\begin{array}{|c|} \hline \img alt="Angry face emoji" data-bbox="353 595 433 675"/> \\ \hline \end{array}, \begin{array}{|c|} \hline \img alt="Angry face emoji" data-bbox="453 595 523 675"/> \\ \hline \end{array} \right), \left(\begin{array}{|c|} \hline \img alt="Angry face emoji" data-bbox="593 595 673 675"/> \\ \hline \end{array}, \begin{array}{|c|} \hline \img alt="Poop emoji" data-bbox="693 595 763 675"/> \\ \hline \end{array} \right) \right\}$$

So $F \times N$ is a face emoji paired with a negative emoji.

Remark. $F \times N \neq N \times F$.

Application: Database management

One application of set theory in computer science is in **database management systems** (DBMS), particularly in **SQL query optimization**.

Example

Imagine a company has two database tables:

- **Customers** (Set C): Contains all registered customers.
- **Orders** (Set O): Contains all customer orders.

A database query like:

```
SELECT * FROM Customers
WHERE CustomerID NOT IN (SELECT CustomerID FROM Orders);
```

is using the set difference $C - O$ to find customers who have never placed an order.

Key Set Operations in Databases

- Union ($A \cup B$) $\rightarrow$ UNION in SQL (Combining results from multiple queries)
- Intersection ($A \cap B$) $\rightarrow$ INTERSECT in SQL (Finding common data between queries)
- Set Difference ($A - B$) $\rightarrow$ EXCEPT or NOT IN in SQL (Finding items in one table but not the other)

Random group assignments

Ashley, Isabelle: 4

Bailey, Tanna: 9

Blaisdell, Olyver: 11

Brown, Ashton: 6

Brown, Jonathan: 10

Cardenas, Dominic: 13

Cutler, George: 13

Fabik, Roland: 3

Fellows, Audrey: 7

Fitzgerald, Ryan: 10

Fuller, Ryan: 7

Griffen, Clara: 10

Grutsch, Tanner: 1

Hager, Nathan: 2

Horner, Jayden: 5

Johnson, Jaret: 8

Kintzel, Samantha: 5

Lameres, Alexis: 4

Lee, Holden: 8

Mayala, Reece: 7

McLean, Connor: 5

Mcbride, Daniel: 13

Miller, Maverick: 11

Moss, Emily: 1

Mousseau, Matthew: 9

Neuman, Evan: 9

Putnam, John: 6

Ramos, Vydel: 6

Rathbun, Jedidiah: 8

Roberts, Matthew: 12

Roller, Nathan: 3

Ryan, Otis: 3

Shevtsov, Timothy: 2

Sinclair, Aidan: 2

Smith, Charles: 12

Stensland, Emma: 4

Thiessen, Logan: 12

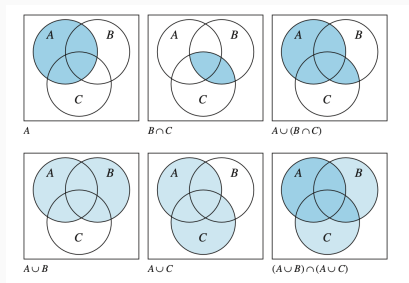
Thompson, Jaiden: 1

Vath, Ethan: 11

Group Exercises: Set Operations

1. Let $A = \{1, 2, 3, 4, 5\}$ and $B = \{4, 5, 6, 7\}$. Please compute (a) $A \cup B$, (b) $A \cap B$, (c) $A - B$, (d) $B - A$, (e) $A \triangle B$, (f) $A \times B$, and (g) $B \times A$.
2. The distributive properties for sets are
$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C), \quad \text{and} \quad A \cap (B \cup C) = (A \cap B) \cup (A \cap C).$$

The Venn diagram below illustrates the first distributive property. Make a Venn diagram to illustrate the second distributive property.



3. Prove that the second identity in #2 holds. *Hint:* Use Theorem 7.2 (Boolean algebra properties).
4. How many integers in the range 1 to 1000 (inclusive) are divisible by 2 or by 5? Justify your answer. *Hint:* Use Proposition 12.4 (Size of a Union).

Solution to Group Exercise #1 (a-e)

Problem. Let $A = \{1, 2, 3, 4, 5\}$ and $B = \{4, 5, 6, 7\}$. Please compute (a.) $A \cup B$, (b.) $A \cap B$, (c.) $A - B$, (d.) $B - A$, and (e.) $A \triangle B$.

Solution.

(a.) $A \cup B = \{1, 2, 3, 4, 5, 6, 7\}$.

(b.) $A \cap B = \{4, 5\}$.

(c.) $A - B = \{1, 2, 3\}$.

(d.) $B - A = \{6, 7\}$.

(e.) $A \triangle B = \{1, 2, 3, 6, 7\}$. [Note: $A \triangle B = (A - B) \cup (B - A)$.]

Solution to Group Exercise #1 (f-g)

Problem. Let $A = \{1, 2, 3, 4, 5\}$ and $B = \{4, 5, 6, 7\}$. Please compute (f.) $A \times B$ and (g.) $B \times A$.

Solution.

(f.)

$$A \times B = \left\{ (1, 4), (1, 5), (1, 6), (1, 7), \right. \\ (2, 4), (2, 5), (2, 6), (2, 7), \\ (3, 4), (3, 5), (3, 6), (3, 7), \\ (4, 4), (4, 5), (4, 6), (4, 7), \\ \left. (5, 4), (5, 5), (5, 6), (5, 7) \right\}$$

(g.)

$$B \times A = \left\{ (4, 1), (4, 2), (4, 3), (4, 4), (4, 5) \right. \\ (5, 1), (5, 2), (5, 3), (5, 4), (5, 5) \\ (6, 1), (6, 2), (6, 3), (6, 4), (6, 5) \\ \left. (7, 1), (7, 2), (7, 3), (7, 4), (7, 5) \right\}$$

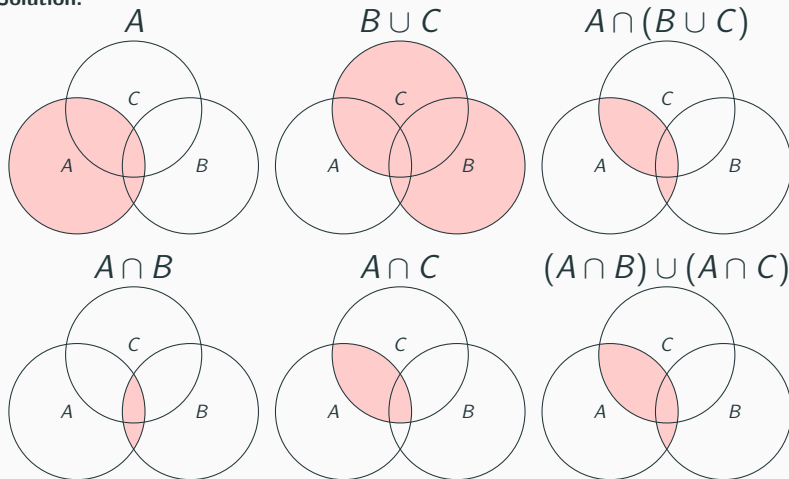
Solution to Group Exercise #2

Problem. The distributive properties for sets are

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C), \quad \text{and} \quad A \cap (B \cup C) = (A \cap B) \cup (A \cap C).$$

Make a Venn diagram to illustrate the second distributive property.

Solution.



Solution to Group Exercise #3

Problem. The distributive properties for sets are

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C), \quad \text{and} \quad A \cap (B \cup C) = (A \cap B) \cup (A \cap C).$$

Prove that the second identity above holds. Hint: You may want to refer back to Theorem 7.2.

Solution to Group Exercise #3

Problem. The distributive properties for sets are

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C), \quad \text{and} \quad A \cap (B \cup C) = (A \cap B) \cup (A \cap C).$$

Prove that the second identity above holds. Hint: You may want to refer back to Theorem 7.2.

Solution.

$$A \cap (B \cup C) = \{x : x \in A \text{ and } x \in B \cup C\} \quad (\text{by def. intersection } \cap)$$

Solution to Group Exercise #3

Problem. The distributive properties for sets are

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C), \quad \text{and} \quad A \cap (B \cup C) = (A \cap B) \cup (A \cap C).$$

Prove that the second identity above holds. Hint: You may want to refer back to Theorem 7.2.

Solution.

$$\begin{aligned} A \cap (B \cup C) &= \{x : x \in A \text{ and } x \in B \cup C\} && \text{(by def. intersection } \cap) \\ &= \{x : x \in A \text{ and } (x \in B \text{ or } x \in C)\} && \text{(by def. union } \cup) \end{aligned}$$

Solution to Group Exercise #3

Problem. The distributive properties for sets are

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C), \quad \text{and} \quad A \cap (B \cup C) = (A \cap B) \cup (A \cap C).$$

Prove that the second identity above holds. Hint: You may want to refer back to Theorem 7.2.

Solution.

$$\begin{aligned} A \cap (B \cup C) &= \{x : x \in A \text{ and } x \in B \cup C\} && \text{(by def. intersection } \cap) \\ &= \{x : x \in A \text{ and } (x \in B \text{ or } x \in C)\} && \text{(by def. union } \cup) \\ &= \{x : (x \in A \text{ and } x \in B) \text{ or } (x \in A \text{ and } x \in C)\} && \text{(by distributive prop.; Thm 7.2)} \end{aligned}$$

Solution to Group Exercise #3

Problem. The distributive properties for sets are

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C), \quad \text{and} \quad A \cap (B \cup C) = (A \cap B) \cup (A \cap C).$$

Prove that the second identity above holds. Hint: You may want to refer back to Theorem 7.2.

Solution.

$$\begin{aligned} A \cap (B \cup C) &= \{x : x \in A \text{ and } x \in B \cup C\} && \text{(by def. intersection } \cap) \\ &= \{x : x \in A \text{ and } (x \in B \text{ or } x \in C)\} && \text{(by def. union } \cup) \\ &= \{x : (x \in A \text{ and } x \in B) \text{ or } (x \in A \text{ and } x \in C)\} && \text{(by distributive prop.; Thm 7.2)} \\ &= \{x : (x \in A \cap B) \text{ or } (x \in A \cap C)\} && \text{(by def. intersection } \cap) \end{aligned}$$

Solution to Group Exercise #3

Problem. The distributive properties for sets are

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C), \quad \text{and} \quad A \cap (B \cup C) = (A \cap B) \cup (A \cap C).$$

Prove that the second identity above holds. Hint: You may want to refer back to Theorem 7.2.

Solution.

$$\begin{aligned} A \cap (B \cup C) &= \{x : x \in A \text{ and } x \in B \cup C\} && \text{(by def. intersection } \cap) \\ &= \{x : x \in A \text{ and } (x \in B \text{ or } x \in C)\} && \text{(by def. union } \cup) \\ &= \{x : (x \in A \text{ and } x \in B) \text{ or } (x \in A \text{ and } x \in C)\} && \text{(by distributive prop.; Thm 7.2)} \\ &= \{x : (x \in A \cap B) \text{ or } (x \in A \cap C)\} && \text{(by def. intersection } \cap) \\ &= (A \cap B) \cup (A \cap C) && \text{(by def. union } \cup) \end{aligned}$$

Solution to Group Exercise #3

Problem. The distributive properties for sets are

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C), \quad \text{and} \quad A \cap (B \cup C) = (A \cap B) \cup (A \cap C).$$

Prove that the second identity above holds. Hint: You may want to refer back to Theorem 7.2.

Solution.

$$\begin{aligned} A \cap (B \cup C) &= \{x : x \in A \text{ and } x \in B \cup C\} && \text{(by def. intersection } \cap) \\ &= \{x : x \in A \text{ and } (x \in B \text{ or } x \in C)\} && \text{(by def. union } \cup) \\ &= \{x : (x \in A \text{ and } x \in B) \text{ or } (x \in A \text{ and } x \in C)\} && \text{(by distributive prop.; Thm 7.2)} \\ &= \{x : (x \in A \cap B) \text{ or } (x \in A \cap C)\} && \text{(by def. intersection } \cap) \\ &= (A \cap B) \cup (A \cap C) && \text{(by def. union } \cup) \end{aligned}$$

Remark. We used the distributive property for *Boolean Algebra* (from Theorem 7.2) to prove the distributive property for *sets*.

Solution to Group Exercise #4

Problem. How many integers in the range 1 to 1000 (inclusive) are divisible by 2 or by 5? Justify your answer.

Sketch of Solution. Let

$$A := \{x \in \mathbb{Z} : 2|x\}$$

$$B := \{x \in \mathbb{Z} : 5|x\}$$

We want $|A \cup B|$. That's hard to calculate.

However, we have

$$|A| = 500 \quad (\text{by skip counting})$$

$$|B| = 200 \quad (\text{by skip counting})$$

$$|A \cap B| = \left| \{x \in \mathbb{Z} : 10|x\} \right| = 100 \quad (\text{by skip counting})$$

Now we apply the identity to find

$$\begin{aligned} |A \cup B| &= |A| + |B| - |A \cap B| \\ &= 500 + 200 - 100 = 600. \end{aligned}$$